

SMART INDIA HACKATHON 2026



TITLE PAGE

VEIL: Evidence-first investigation workspace

Problem Statement ID: [TO FILL]

Problem Statement Title: [TO FILL]

Theme: [TO FILL]

PS Category: Software

Team ID: [TO FILL]

Team Name: [REGISTERED NAME]



Your
Team
Name

IDEA TITLE



Proposed Solution (Describe your Idea/Solution/Prototype)

VEIL connects separate records into entities, relationships and explainable leads.

Detailed explanation of the proposed solution

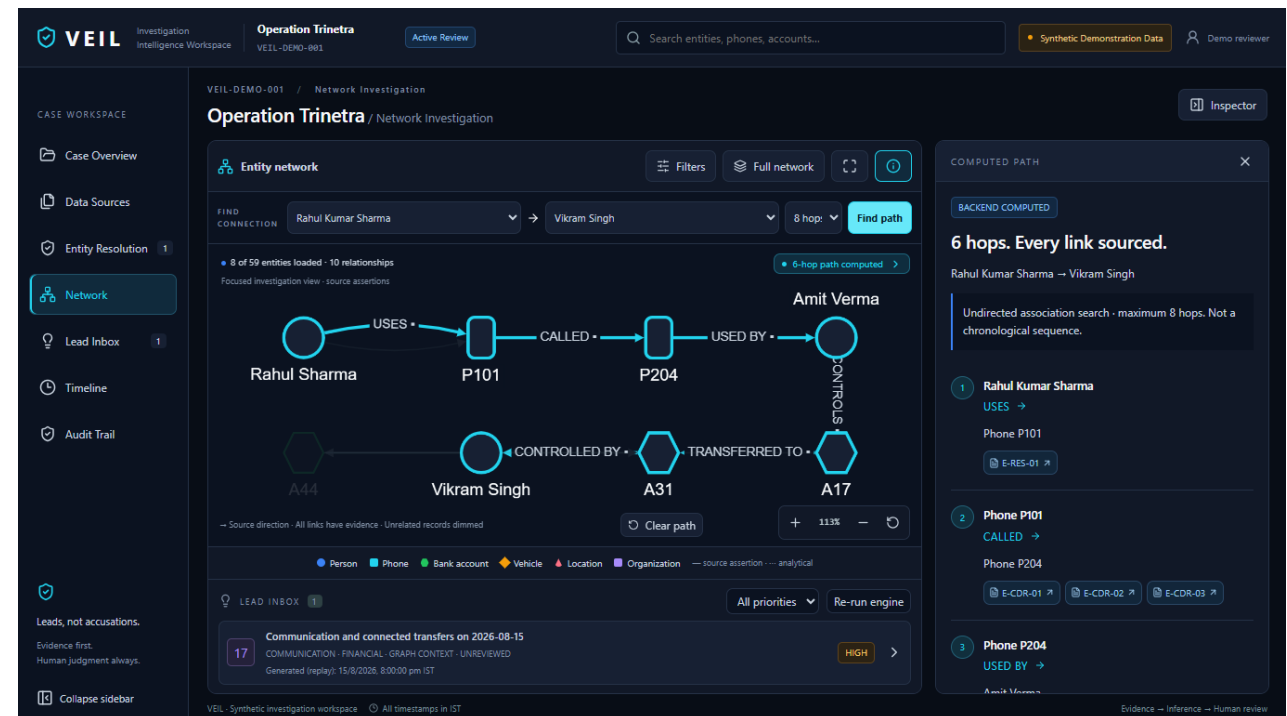
Reports, CDR, transactions and vehicle records in one workspace.

How it addresses the problem

Investigators can follow a connection and open its exact source record.

Innovation and uniqueness of the solution

Cross-source graph + provenance + reversible human review.



Actual prototype: a computed, sourced association path. Synthetic data.

Technologies to be used

React / TypeScript / Cytoscape: investigator interface

Python / FastAPI / NetworkX: APIs and graph analysis; JSON + SQLite: records and decisions

Methodology and process for implementation



ML / NLP

spaCy en_core_web_sm extracts report entities; regex extracts structured identifiers.

Graph algorithms

Bounded shortest paths and connected components expose recorded associations.

Rules + statistical ML

Temporal rules generate leads. Isolation Forest supports secondary anomaly triage.

Human decision

Review uncertain mentions, inspect source evidence, confirm or reject, retain audit history.

Analysis of the feasibility of the idea

Implemented: local frontend + backend, four input types, NER, graph traversal and evidence review.

Verification: 98 backend tests passed on 5 September 2026 (two dependency warnings).

Potential challenges and risks

Synthetic case only; real-world accuracy and throughput remain unmeasured.

NER errors, ambiguous identities, incomplete records and small anomaly cohorts can mislead.

Strategies for overcoming these challenges

Current controls: source provenance, unverified extraction, reversible review and sample guards.

Production roadmap: agency validation, authenticated roles, encryption and protected audit storage.

Scale after profiling: asynchronous ingestion and indexed graph persistence.

Potential impact on the target audience

Investigators can connect scattered records and inspect why a lead deserves review.

Benefits of the solution

Less repeated cross-referencing.

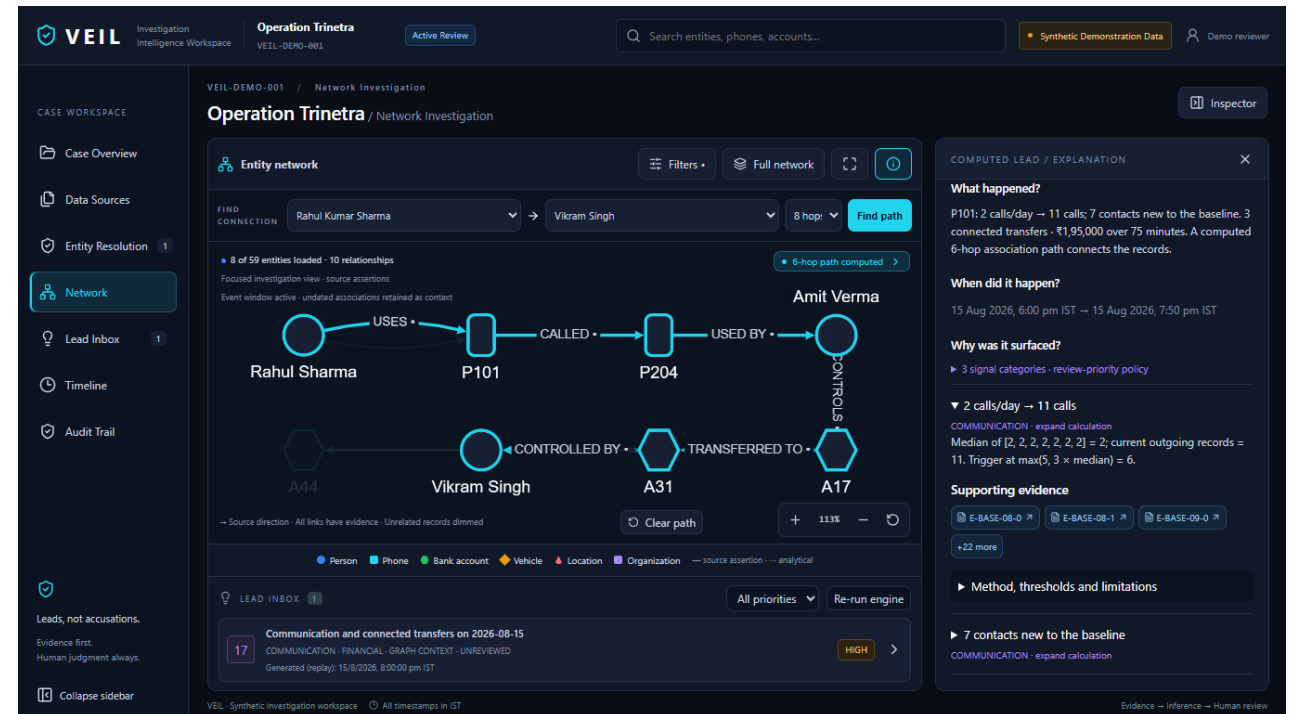
Source-linked explanations.

Reviewable human decisions.

Synthetic demonstration

11 calls vs a 2/day baseline; 3 connected transfers over 75 minutes; a 6-hop path.

Together, these create a review-priority lead.



Expected benefits; no measured time savings or crime-detection claim.

Details / Links of the reference and research work

spaCy English pipeline

Local named-entity extraction; en_core_web_sm

https://spacy.io/models/en#en_core_web_sm

NetworkX shortest paths

Graph traversal methods used by the prototype

https://networkx.org/documentation/stable/reference/algorithms/shortest_paths.html

scikit-learn Isolation Forest

Unsupervised anomaly detection for secondary triage

<https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.IsolationForest.html>

Project evidence and validation

Source: backend/ner_service.py, behavioral_profiler.py, demo_paths.py and lead_engine.py.

Evidence: tests/ (98 passed), artifacts/first-round/api/ and actual prototype screenshots.

Evaluation boundary: synthetic data; operational accuracy requires agency validation.